

Supplementary File S6

Validation and sensitivity analyses for the PESTFLY benchmark

Reader guide to the compressed archive for the population genomic analysis of *Bactrocera dorsalis*. The archive contains seven supplementary analyses, their complete outputs, and baseline query trajectories across the evaluated SNP subsets.

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Benchmark context

The frozen snapshot contains 352 specimens: 330 references and 22 queries. P1 compares 105 African with 215 Asian references. Ten references from Hawaii and Papua New Guinea are outside the represented P1 classes. P2 evaluates four African subregions; P3 evaluates three Asian subregions.

Reading the metrics

Raw accuracy compares the top call with the recorded reference class among evaluable specimens. **Reportability** is the fraction meeting the applicable acceptance rules. **Accuracy among accepted calls** uses accepted calls as its denominator. **Query agreement** compares a query call with its saved baseline call; it does not measure accuracy against known query origin.

Assignments describe genomic affinity to represented classes. They do not establish actual collection origin, a transport pathway, or the presence of the true source in the reference set.

Saved sources. README.md and source_map.tsv give the archive layout and exact paths. Numerical tables remain authoritative; percentages in this guide are rounded to one decimal place.

S6.1 and S6.2

S6.1 Geographic out of scope challenge

Ten geographically unrepresented references, five from Hawaii and five from Papua New Guinea, were excluded from P1 training and treated as challenge queries. All ten received accepted High confidence Asian macroregion affinities. Hawaii samples affiliated with East Asia and Papua New Guinea samples with Southeast Asia.

All ten had final posterior support, posterior gap and global K agreement of 1.0. None was flagged uncertain. These results demonstrate that closed set confidence cannot identify every geographically unrepresented source.

The challenge reused validated existing Step 04 tabular outputs. No challenge workbook was created because the original Excel writer failed on the empty Africa conditional branch. Complete TSV and RDS assignments remain the recorded authoritative inputs.

Saved sources. other_regions_challenge_results.tsv, other_regions_challenge_by_location.tsv and challenge_run_info.rds under results/validation/01_out_of_scope_challenge/.

S6.2 Grouped geographic cross validation

A country or site was held out as a group. Missingness filtering, minor allele count filtering and FST ranking used the remaining training references, within the retained Step 02 SNP candidates.

Panel	Holdout	Evaluable / total	Raw accuracy	Reportable	Accepted accuracy
P1	country	320/320	100.0%	99.4%	100.0%
P1	site	320/320	100.0%	99.4%	100.0%
P2	country	100/105	93.0%	75.0%	93.3%
P2	site	100/105	93.0%	75.0%	93.3%
P3	country	135/215	83.7%	74.1%	92.0%
P3	site	215/215	91.2%	85.6%	95.7%

Five C_Africa references in the Congo country fold and Congo/Kabare site fold were non evaluable because their true class was absent or undersupported after holdout. The China country fold similarly made 80 E_Asia references non evaluable. Rates in the table use evaluable specimens; these unsupported folds are recorded separately.

This validates training based filtering and reranking among retained candidates. It does not repeat ortholog discovery or choose another SNP from each original ortholog. Pooled rates above are distinct from the fold balanced rates also supplied in the archive.

Saved sources. grouped_cv_panel_summary.tsv, grouped_cv_non_evaluable_folds.tsv, grouped_cv_raw.tsv.gz and grouped_cv_params.tsv under results/validation/02_grouped_geographic_cv/.

S6.3 Reference set audit

The audit examines the frozen metadata, reference representation, published specimen cautions, ortholog completeness and genotype similarity. It records evidence and sensitivity tiers without automatically removing specimens from the primary analysis.

Audit measure	Recorded result
Duplicated sample IDs / reference accessions	0 / 0
Critical / high metadata issues	0 / 0
Documentation completeness issues	50
Near duplicate reference pairs	0
References with published cautions	12
References below 95% ortholog coverage	2 of 330

Published specimen cautions

Nine references were published nuclear PCA outliers, including five hybrid candidates. Three additional references had COI taxonomic cautions. These are evidence flags for sensitivity analysis, rather than taxonomic determinations.

Specimen	Country	Published flag categories
CDKB02	Congo	COI taxonomic caution
YNJH01	China	Nuclear PCA outlier
YNJH02	China	Nuclear PCA outlier
YNJH03	China	Nuclear PCA outlier
YNJH05	China	Nuclear PCA outlier
IDJI04	Indonesia	Nuclear PCA outlier; Hybrid candidate
MMYG01	Myanmar	Nuclear PCA outlier; Hybrid candidate
MMYG02	Myanmar	Nuclear PCA outlier; Hybrid candidate
MMYG04	Myanmar	Nuclear PCA outlier; Hybrid candidate
MMYG05	Myanmar	Nuclear PCA outlier; Hybrid candidate
LKAD01	Sri Lanka	COI taxonomic caution
LKAD04	Sri Lanka	COI taxonomic caution

ReuPie1 and ReuPie4 fell below 95% ortholog coverage. Similarity screening used 330 profiles across 11,086 P1 SNPs, a minimum of 1,000 shared SNPs and a near duplicate threshold of 0.995. No qualifying near duplicate, identical on overlap or fully identical pairs were found.

Five panel classes were represented by only one source collection. Six of eight collection holdouts were feasible by class support. Geographic class and source collection can therefore be confounded; feasibility is not a completed collection holdout performance estimate.

Saved sources. REFERENCE_SET_AUDIT_SUMMARY.txt, retained_reference_flag_summary.tsv, reference_ortholog_coverage.tsv and collection_holdout_feasibility.tsv under results/validation/03_reference_set_audit/.

S6.4 Mascarene coding sensitivity

Forty Reunion and five Mauritius references have Asian lineage affinities in the baseline coding. The sensitivity analysis rebuilds panels under three codings: baseline Asia / SE_Asia, geographic Africa / E_Africa, and Other, excluded from the two class P1 comparison.

Query changes relative to baseline

Alternative coding	Macro calls changed	Subregion calls changed	Subregion confidence changed
Geographic Africa	0	2	3
Other / excluded	0	0	1

All 22 macroregion calls and their confidence categories were unchanged in both alternative codings. Some subregion calls or confidence categories changed, as shown above.

Individual fixed panel leave one out results

Coding	Panel	References	Correct	Raw accuracy
Baseline Asia	P1	320	320	100.0%
Baseline Asia	P2	105	103	98.1%
Baseline Asia	P3	215	199	92.6%
Geographic Africa	P1	320	314	98.1%
Geographic Africa	P2	150	146	97.3%
Geographic Africa	P3	170	154	90.6%
Other / excluded	P1	275	275	100.0%
Other / excluded	P2	105	103	98.1%
Other / excluded	P3	170	154	90.6%

The geographic Africa coding moved 45 references between conditional branches. Excluding them reduced the P1 reference set to 275. Differences across scenarios combine changed class definitions, reference composition and reconstructed marker panels.

Each scenario retains metadata changes, panel indexes, marker overlap, assignments across K, and leave one out results. Those results remain conditional on the scenario marker panel; they do not establish a true geographic origin for the Mascarene lineage.

Saved sources. return_bundle/scenario_definitions.tsv, query_assignment_change_summary.tsv and loo_summary_comparison.tsv under results/validation/04_mascarene_coding_sensitivity/. Complete records are under scenarios/.

S6.5 Balanced reference downsampling

Each design samples equal training counts per class and a disjoint reference test set, with 30 replicates and base seed 20260930. There are 16 completed designs and 480 design replicates. Filtering and reranking use training references within the retained Step 02 candidates.

At most five test references per class are available; small classes constrain both training size and test count. P2 permits only three or four training references per class. The table reports replicate means and the actual test count per class.

Panel	Train / class	Test / class	Raw accuracy	Reportable	Uncertain
P1	3	5	100.0%	28.3%	71.7%
P1	5	5	100.0%	64.0%	36.0%
P1	10	5	100.0%	50.7%	49.3%
P1	20	5	100.0%	99.7%	0.3%
P1	40	5	100.0%	100.0%	0.0%
P1	80	5	100.0%	99.7%	0.3%
P1	100	5	100.0%	99.7%	0.3%
P1	104	1	100.0%	98.3%	1.7%
P2	3	2	94.2%	0.0%	100.0%
P2	4	1	95.8%	80.0%	20.0%
P3	3	5	90.9%	0.0%	100.0%
P3	5	5	89.8%	0.0%	100.0%
P3	10	5	91.8%	0.0%	100.0%
P3	20	5	93.1%	92.7%	7.3%
P3	40	5	92.9%	91.6%	8.4%
P3	44	1	88.9%	92.2%	7.8%

P1 mean raw accuracy is 100% at every training size. At three references per class, mean uncertainty is 71.7%; at ten, it is 49.3%. Thus a correct top call does not always meet the cumulative K reporting criteria. Neither raw accuracy nor reportability is assumed to vary monotonically with training size.

Replicate quantiles, class specific results, query stability, training and test membership, marker overlap and individual test predictions are retained. Test sets are disjoint from training within a replicate; references can recur across replicates.

Saved sources. downsampling_design.tsv, downsampling_performance_summary.tsv, downsampling_params.tsv and FIG_reference_downsampling_sensitivity.pdf under results/validation/05_reference_downsampling/.

S6.6 Alternative marker set resampling

For each panel and K of 500, 1,000, 2,000 or 5,000, 15 rank matched A/B partitions of the existing top 2K ranking produce 30 alternative marker sets. The archive contains 12 panel by K designs and 360 alternative panels. Base seed: 20261001.

A and B are disjoint within a partition and each contains one SNP per ortholog, with half its markers from the conventional Top K set. Sets from different partitions can overlap. All recorded complement and one SNP per ortholog checks passed.

Panel	K	Mean LOO raw accuracy	Mean query agreement with primary call
P1	500	99.4%	86.4%
P1	1,000	99.4%	95.5%
P1	2,000	100.0%	95.5%
P1	5,000	100.0%	100.0%
P2	500	92.4%	62.8%
P2	1,000	94.4%	67.5%
P2	2,000	95.9%	74.7%
P2	5,000	97.8%	89.4%
P3	500	95.9%	96.2%
P3	1,000	94.6%	99.5%
P3	2,000	93.1%	99.5%
P3	5,000	92.2%	100.0%

The query comparison includes 22 P1, 12 P2 and seven P3 queries. Agreement with primary calls evaluates sensitivity to the marker set. The true geographic origin of the queries is not known from this comparison, so agreement is not an accuracy measure.

Fixed K acceptance uses posterior support, posterior gap and usable SNP thresholds. It does not apply the cumulative K convergence rules used in the primary multi K workflow. Stability is assessed across alternative sets, conditional on the existing ranking.

The archive retains marker membership, pairwise overlap, predictions, per class results, set performance, fixed Top K baselines and the summary figure. These are alternative marker panels, rather than independent repetitions of ortholog or SNP discovery.

Saved sources. marker_resampling_design.tsv, marker_resampling_loo_performance_summary.tsv, marker_resampling_query_overall_summary.tsv and FIG_marker_resampling_sensitivity.pdf under results/validation/06_marker_resampling/.

S6.7 Reference exclusion sensitivity

The primary analysis retains all references. Three sensitivity scenarios exclude 12 published caution specimens, two low coverage specimens, or the combined 14. Each scenario rebuilds panels and repeats assignments and individual fixed panel leave one out validation.

Scenario	Panel	References	Correct	Raw accuracy
Primary	P1	320	320	100.0%
Primary	P2	105	103	98.1%
Primary	P3	215	199	92.6%
Published cautions	P1	308	308	100.0%
Published cautions	P2	104	101	97.1%
Published cautions	P3	204	187	91.7%
Low coverage	P1	318	318	100.0%
Low coverage	P2	105	103	98.1%
Low coverage	P3	213	197	92.5%
Combined	P1	306	306	100.0%
Combined	P2	104	101	97.1%
Combined	P3	202	185	91.6%

Query changes relative to the primary analysis

Exclusion	Subregion calls changed	Subregion confidence changed
Published cautions	2	5
Low coverage	0	1
Combined	2	6

All 22 macroregion calls and confidence categories were unchanged. P1 raw leave one out accuracy remained 100% at reference counts of 320, 308, 318 and 306. Changes in conditional subregion calls and confidence remain visible in the comparison tables.

Exclusion is a sensitivity analysis of reference evidence and completeness. It does not retrospectively certify flagged specimens as invalid. Complete scenario metadata, markers, trajectories, leave one out tables and parameter records are retained.

Saved sources. `return_bundle/reference_exclusion_evidence.tsv`, `query_assignment_change_summary.tsv` and `loo_summary_comparison.tsv` under `results/validation/07_exclusion/`. Complete records are under `scenarios/`.

Complete query trajectories

Baseline Step 04 retains every evaluated Top K row, including low K rows with insufficient usable SNPs or unavailable posterior values. There are 656 rows, with 16 K values per evaluated query branch.

Branch	Query specimens	Raw table rows	All SNP K
P1 Africa / Asia	22	352	11,086
P2 African subregion	12	192	11,001
P3 Asian subregion	7	112	11,086

The common K grid is 1, 2, 3, 4, 5, 10, 20, 50, 100, 200, 500, 1,000, 2,000, 5,000 and 10,000, followed by the all SNP endpoint shown above. K denotes the requested ranked subset; n_snps_used records the usable loci for that specimen.

Retained fields and conditional branches

Raw tables contain panel_id, group_col, K, sample_id, top_group, top_posterior, second_group, second_posterior, posterior_gap, n_snps_used, confidence, status and note. They retain top and second class support, rather than a full posterior distribution over every class.

Subregion evaluation is conditional on an accepted macroregion assignment. Twelve queries enter P2 and seven enter P3. Three queries with failed macroregion reporting criteria have no subregion branch; this is a defined workflow outcome, rather than an omitted trajectory.

Six baseline reporting criteria

Criterion	Archived requirement
Usable SNP count at final usable K	At least 200 for P1; 300 for P2 / P3
Posterior support	At least 0.85
Posterior gap	At least 0.20 for P1; 0.25 for P2 / P3
Global agreement over usable K	At least 0.90
Agreement in the largest half of usable K	1.00; tail size uses the ceiling
Number of usable K values	At least six

All six requirements must pass. Posterior support of at least 0.95 gives High confidence; 0.85 to below 0.95 gives Moderate when all requirements pass. The largest $\text{ceil}(n_{\text{usable_K}} \times 0.5)$ K values define the stability tail.

The native Step 07 report records 19 macroregion PASS and three FAIL outcomes, plus 12 subregion PASS, seven FAIL and three NOT_EVALUATED outcomes. Original final classes and confidence labels are preserved. The reporting checks and formatter passed in R 4.5.1.

Saved sources. P1_macroregion_raw.tsv, P2_africa_subregion_raw.tsv, P3_asia_subregion_raw.tsv, corresponding stability tables, final_assignment.tsv, step4_K_grid_used.tsv and step4_params.tsv under results/04_origin_assignment/.

Implementation and reproduction

Archived estimator and MI checks

Step 02 uses binary consensus allele calls with ambiguous calls treated as missing. Hudson sample sizes are counts of non missing consensus calls. Finite negative locus estimates are retained; undefined denominators, insufficient calls and invalid inputs produce unavailable estimates.

The estimator numerator is $(p1 - p2)^2 - p1(1 - p1)/(n1 - 1) - p2(1 - p2)/(n2 - 1)$. Its denominator is $p1(1 - p2) + p2(1 - p1)$. Four archived test fixtures expect 1, $-1/3$, unavailable and $-2/3$, with group swap and allele flip symmetry checks.

The Step 02 run record reports the FST self test as PASS. The Step 03 run record reports successful MI self tests: zero entropy with unused factor levels, one bit for perfect balanced binary association, and zero for balanced independence. Every reported locus was checked for finite, nonnegative MI no greater than target entropy. MI is diagnostic and does not select markers or change the assignment model.

Saved sources. results/02_snp_panels/step3_run_info.rds and results/03_mi_diagnostics/step03_run_info.rds.

Checks that can run without sequence inputs

The new isolated implementation wrapper evaluates only the named function definitions parsed from Steps 02 and 03. It also checks missing inputs and insufficient calls. The wrapper and Step 07 reporting checks use base R.

```
Rscript tools/check_implementation.R
Rscript steps/07_authority_facing_report/check_reporting.R
```

The isolated implementation wrapper, reporting checks and revised Step 07 formatter passed in the supplied Windows R 4.5.1 session on 5 October 2026. Native TSV, workbook and RDS exports were independently verified. No statistical model or sequence analysis was rerun to assemble S6.

Resources, versions and integrity

R 4.5.1 and recorded dependency versions are listed in docs/INSTALL.md and environment/benchmark_versions.tsv. Panel based analyses 01, 02, 03, 05 and 06 use supplied metadata, QC and SNP resources. Rebuilding scenario panels in analyses 04 and 07 also requires regenerated FASTA. The separately distributed PHYLIP inputs are not included in S6.

Use a separate copy for recomputation because default outputs may be replaced. A complete transitive dependency lock has not been reconstructed. Native reporting verification is documented in docs/NATIVE_R_VERIFICATION.md.

```
python3 verify_archive.py
```

The verifier checks every recorded file size and SHA256. The archive manifest excludes itself. Only byte identical summary export copies with retained counterparts in the same analysis are omitted; duplicate_exports.tsv records their mappings. All scientifically distinct outputs remain included.

Historical run records retain original paths and labels. Path and text edits are recorded in docs/location_index_changes.tsv and docs/textual_output_changes.tsv. Scientific outputs are preserved. Code uses MIT; public documentation and results use CC BY 4.0. Copyright 2026 Massimiliano Virgilio and Wannes Dermauw. Alignments are distributed upon request and excluded from S6. Raw data deposit: <https://doi.org/10.5281/zenodo.20340447> (files restricted). Workflow archive: <https://doi.org/10.5281/zenodo.23187057>. See ALIGNMENT_ACCESS.md for access requests.